

# RBF-FD Solver: Least Squares Method for Operator Reproduction

Dennis Corraliza

9/2/2026

# 1 Setup:

Standard RBF-FD methods assume a native space for the solution set of a PDE problem in form of:

$$-Lu = -\nabla \cdot (A\nabla u) = f \quad \text{in } \Omega, \quad (1)$$

$A$  is an symmetric positive definite anisotropic tensor and  $f$  is the forcing term acting in the domain  $\Omega$ . The problem is subject to Neumann and Dirichlet Boundary Conditions partitioned within  $\partial\Omega$  of the form:

$$u = g_D \quad \text{on } \Gamma_D \subset \partial\Omega, \quad (2)$$

$$\frac{\partial u}{\partial n} \equiv n \cdot A\nabla u = g_N \quad \text{on } \Gamma_N \subset \partial\Omega, \quad (3)$$

Rather than assuming a form for the solution, our goal is to approximate the operator through weights dual to the  $u$  sampled at nodes. We define the radial basis functions in the form  $\phi_i(\mathbf{x}) := \phi(\|\mathbf{x} - \mathbf{y}_i\|)$  for  $i = 1, \dots, m$  and centers  $\{\mathbf{y}_i\}_{i=1}^m \subseteq \Omega$ . Define the vector space  $V_Y = \text{span}\{\phi_1, \dots, \phi_m\}$ . Let  $X = \{\mathbf{x}_1, \dots, \mathbf{x}_n\} \subseteq \Omega$  be a set of stencil nodes in  $\Omega$ . For the following procedure, assume  $m > n$ , thus more center nodes than stencil nodes per stencil. We can now define  $\Phi \in \mathbb{R}^{n \times m}$ , where  $\Phi_{ij} = \phi_j(\mathbf{x}_i)$ . The role of these radial basis functions will be to approximate the operator  $L$ . Rather than assuming exactness of the form  $L\phi_j(\mathbf{x}) = \sum_i^n w_i(\mathbf{x})\phi_j(\mathbf{x}_i)$  for  $j = 1, \dots, m$  as in the classical RBF-FD, we define a residual function:

$$r_j(\mathbf{x}) := L\phi_j(\mathbf{x}) - \sum_{i=1}^n w_i(\mathbf{x})\phi_j(\mathbf{x}_i) \quad (4)$$

$$r(\mathbf{x}) = L\phi(\mathbf{x}) - \Phi^T \mathbf{w}(\mathbf{x}), \quad \mathbf{w} : \mathbb{R}^n \rightarrow \mathbb{R}^n \text{ for } \mathbf{x} \in \Omega. \quad (5)$$

Suppose  $\{p_i\}_{i=1}^l$  is a set of polynomials satisfying the nullspace condition for the radial basis functions. Let  $\Pi_k = \text{span}\{p_i\}_{i=1}^l$  with polynomial degree up to  $k$ . We want to reproduce the operator exactly on the polynomial basis:

$$Lp_j(\mathbf{x}) := \sum_{i=1}^n w_i(\mathbf{x})p_j(\mathbf{x}_i) \text{ for } j = 1, \dots, l \quad (6)$$

$$Lp(\mathbf{x}) = \mathbf{P}^T \mathbf{w} \quad (7)$$

where  $\mathbf{P}_{ij} := p_j(\mathbf{x}_i) \in \mathbb{R}^{n \times l}$ . Thus, we have a constrained least squares problem in the form:

$$\begin{cases} \min J(\mathbf{w}(\mathbf{x})) := \min \frac{1}{2} \|\mathbf{r}(\mathbf{x})\|_2^2 & \text{for } \mathbf{x} \in \Omega \\ \text{subject to } \Phi^T \mathbf{w} = Lp \end{cases} \quad (8)$$

We can solve this setup by defining the Lagrangian  $\mathcal{L}$  for the problem as follows:

$$\begin{aligned} \mathcal{L}(\mathbf{w}, \boldsymbol{\nu}) &:= J(\mathbf{w}) + \langle \boldsymbol{\nu}, \mathbf{P}^T \mathbf{w} - Lp(\mathbf{x}) \rangle \\ &= \frac{1}{2} \langle \Phi^T \mathbf{w}, \Phi^T \mathbf{w} \rangle - \langle \Phi^T \mathbf{w}, L\phi \rangle + \frac{1}{2} \langle L\phi, L\phi \rangle + \langle \boldsymbol{\nu}, \mathbf{P}^T \mathbf{w} - Lp \rangle \end{aligned}$$

The resulting KKT equations for this system are then:

$$\nabla_{\mathbf{w}} \mathcal{L} = 0 : \Phi L\phi = \Phi \Phi^T \mathbf{w} + \mathbf{P}\boldsymbol{\nu} \quad (9)$$

$$\nabla_{\boldsymbol{\nu}} \mathcal{L} = 0 : \mathbf{P}^T \mathbf{w} = Lp \quad (10)$$

The augmented matrix then takes the form:

$$\begin{bmatrix} \Phi\Phi^T & P \\ P^T & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{w} \\ \boldsymbol{\nu} \end{bmatrix} = \begin{bmatrix} \Phi L\phi \\ Lp \end{bmatrix} \quad (11)$$

The primary goal is to approximate  $Lu(\mathbf{x}) \approx \mathbf{w}^T(\mathbf{x})\mathbf{u}$ , with  $\mathbf{u} = [u(\mathbf{x}_1), \dots, u(\mathbf{x}_n)]^T$ . Define  $L_h u(\mathbf{x}) := \mathbf{w}^T \mathbf{u}$ . Thus, we have the following:

$$\begin{aligned} L_h u(\mathbf{x}) &= \mathbf{w}^T(\mathbf{x})\mathbf{u} \\ &= [\mathbf{w}^T \quad \boldsymbol{\nu}^T] \begin{bmatrix} \mathbf{u} \\ \mathbf{0} \end{bmatrix} \\ &= \left( \begin{bmatrix} \Phi\Phi^T & P \\ P^T & \mathbf{0} \end{bmatrix}^{-1} \begin{bmatrix} \Phi L\phi \\ Lp \end{bmatrix} \right)^T \begin{bmatrix} \mathbf{u} \\ \mathbf{0} \end{bmatrix} \\ &= [L\phi^T \Phi^T \quad Lp^T] \begin{bmatrix} \Phi\Phi^T & P \\ P^T & \mathbf{0} \end{bmatrix}^{-T} \begin{bmatrix} \mathbf{u} \\ \mathbf{0} \end{bmatrix} \\ &= [L\phi^T \Phi^T \quad Lp^T] \begin{bmatrix} \mathbf{q} \\ \boldsymbol{\alpha} \end{bmatrix} \\ &= L\phi^T(\mathbf{x}) (\Phi^T \mathbf{q}) + Lp^T(\mathbf{x}) \boldsymbol{\alpha} \end{aligned}$$

Next, we can show that the coefficients  $\mathbf{q}, \boldsymbol{\alpha}$  relate to the coefficients of  $u$  as follows:

$$\begin{bmatrix} \mathbf{u} \\ \mathbf{0} \end{bmatrix} = \begin{bmatrix} \Phi\Phi^T & P \\ P^T & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{q} \\ \boldsymbol{\alpha} \end{bmatrix} \quad (12)$$

Define  $V_h = \{\rho(x) : \rho(x) = \sum_{i=1}^m \lambda_i \phi_i(x) \text{ such that } \lambda \in \text{Range}(\Phi^T)\} \subseteq V_Y$ . Let  $s \in V_h + \Pi_k$ .